An analysis of the effect of monetary policy changes

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Abstract:

One important role of the Reserve Bank of India or any central bank is to ensure economic stability in the country. For the purpose, the central bank adopts various measures to ensure that the inflation rates, GDP, interest rates, exchange rates, money supply and other target macroeconomic parameters remain under control. It uses reserve ratios like Cash Reserve Ratio and interest rates like Repo rates to control liquidity and inflation in the country. The effectiveness of such policy and monetary rates in ensuring economic stability needs to be verified and tested. The decision maker needs to understand the effect of these changes on the affected (targeted) variable. This research is a part of the working paper on this theme and an attempt to test and verify the effectiveness of the changes in monetary and policy rates on the desired critical variables. What is the effect and how much is the effect? These are the two elementary questions to be answered in the research.

The research uses basic statistical tools such as correlation, regression and advance statistical tools such co-integration to study the effect and draws conclusion based on the results. The data used is from Indian economy and the time period used is 2011-2014. Monthly and quarterly data has been used as required. The research is aimed to provide inputs to decision makers in formulating policies and contribute to the existing literature on the subject.

JEL Codes: G21, E1, E5

Introduction:

‘The biggest challenge facing the conduct of fiscal and monetary policy in India is to continue the accelerated growth process while maintaining price and financial stability’, Rakesh Mohan (2008).

Hutchison et al.(2012) focused on accumulation of international reserves and sterilization by the RBI using quarterly data from 1996 to 2009.Their analysis confirm that an increase in financial integration has changed the policy trade-offs facing emerging market economies like India. Mohan (2008) in his paper focuses on the role of fiscal and monetary policies in the evolution of the Indian economy over the years, with particular attention being given to the reforms undertaken in these policies since the early 1990s. He argues that monetary policy aims to maintain a judicious balance between price stability and economic growth. Palakkeel (2007) argues that apart from interest rate spread and financial deregulation, income, previous periods financial savings and inflation rate can affect compositional variation of financial assets. Gottschalk and Moore (2001) studied the efficacy of inflation targeting regime for Poland and argued that inflation targeting regime could be successful in Poland with understanding of the linkages between monetary policy and inflation outcomes with a focus on prerequisites. Khatkhate (2006) asserted that inflation targeting might be good policy framework for India as RBI always has to be on alert to maintain its credibility and authority in controlling inflation although source may often be non-monetary. The interest rate pass through process is one in which bank interest rates respond to changes in monetary policy rates. This process is simply the rate or process at which the official interest rate is transmitted to other interest rates (Kovanen, 2011). A weak and incomplete interest rate pass-through is an indication of an unhealthy financial system (Aydim, 2007) and the failure of monetary policy to stabilize macroeconomic shocks (Marotta, 2009). In order to tighten liquidity in the market, central bank interest rates are increased which further cuts back on investments by increasing cost of borrowing for banks as well as consumers. It is imperative to study the degree and speed of this change amongst interest rates. In the study conducted by Sander and Kleimerier (2006), it was found that there exists a greater response to anticipated monetary policy changes measures rather than to unanticipated changes. Aziakpono, Wilson and Manuel (2007), found market interest rates to respond quickly to monetary policy rate, the study conducted by Aziakpono and Wilson (2010) found that commercial banks lending rates are more rigid in response to positive shocks in monetary policy official rate in South Africa. *In one of the latest study by Kelulime, 2014, a similar data set was used by where monthly data was used for time period 2007-2012 for regression analysis on macroeconomic factors.* Suthar (2008) observed that the monetary policy intentions depicted by the bank rate of the RBI, the short-term and long-term domestic interest differentials and interest yield differentials, and the rate of change of foreign exchange reserves have a significant impact on the monthly average of the exchange rate between the rupee and the dollar indicating that the exchange rates are affected by monetary policy rates. *Mishra (2013) attempted to evaluate the independence of monetary policy in India from all the possible spheres and concludes that the inflation targeting framework may be amended and targeting inflation band rather than inflation point would be a better option for RBI.*

Research Methodology:

Objective: The general objective is to understand the interrelationship amongst selected macroeconomic variables which are prominent in banking sector, for effective policy making. The specific objective is to understand the dynamics between monetary policy rates and ratios with inflation, liquidity and foreign exchange. The three key functions of the Reserve Bank of India (RBI) are controlling inflation, liquidity management and stabilizing foreign exchange rate. RBI has been using the policy rates (Repo, Reverse Repo, Bank Rate) and statutory ratios (CRR,SLR) to tame inflation and control liquidity in the economy but the effectiveness of this approach has always been questioned. This paper tries to understand this dynamics between monetary policy, inflation and liquidity by working on mathematical models and drawing conclusions based on the analysis.

Interest rate pass through process has been highlighted in previous researches indicating the effect of one rate over another in an economic system (Kovanen 2011,Kelilume 2014). References were found for and against , both, in using the level data (Ahmad and Rao,2006) or first difference of data for analysis(Kelilume,2014) and this given study used both types of data for analysis. Quarterly data for GDP has been used in the study as this data set is available from RBI (Prabheesh et al,2006). This study also used quarterly data for GDP and related varaibales.

The data used in the research is for the time period January, 2011 to December, 2013. The raw data was retrieved from the RBI database. The variables used in the study are given below along with a brief description of the same. The terms in bracket indicate the acronym used in equations and SPSS analysis. Also ‘Gretl’ software was used as an additional tool for analysis.

Table 1: List of variables used along with their descriptions

1. Cash Reserve Ratio (CRR): The percent of deposits which banks have to keep with RBI as cash.
2. Repo rates (Repo): The rate of interest at which banks borrow from RBI.
3. Yields on Bonds(BondYld):The yield to maturity of 10 years government bonds.
4. Exchange Rate of one US Dollar in Indian rupees (USD.INR)
5. Call money interest rates(CallMny):The rate of interest applicable on call money
6. Consumer Price Index(CPI): The measure of inflation applicable on consumer prices
7. Deposits of the banking sector (Dep)
8. Borrowings of banking sector from RBI (Brwg.RBI)
9. Balance of Banks with RBI(Bal.RBI)
10. Banking credit (Bnk.Credt): The level of credit extended by banks
11. Statutory Liquidity Ratio (SLR): The percent of deposits which the banks are required to invest in liquid assets, excluding CRR.
12. Reverse Repo Rate(ReRepo): The rate at which the banks park their money with RBI
13. Bank Rate (BnkRate): The long term reference rate as decided by RBI
14. Velocity of Money(VoM): Calculated as a percentage of GDP over Broad money supply
15. Money Demand (MD): as a function of velocity of money, GDP, Inflation and interest rates.
16. GDP: The economic output of the country taken as current price at base 2011-12

The variables used have been categorized as:

Input variables: Cash Reserve Ratio (CRR), Statutory Liquidity Ratio (SLR),Repo rates , Reverse Repo rates and Bank Rates

Output variables: Foreign Exchange Rate, Consumer Price Index (CPI), Deposits of the banking sector, Banking credit, GDP, Money Demand and Velocity of Money

The units of various parameters were different for different variables, varying from percentages to absolute values to index numbers. For the convenience of effective analysis, data of all the variables was worked upon for uniformity. All the variables were converted into index numbers with their initial value of the time period made as the base value at 100 and the subsequent values were converted into index number at this base. Thus, the size effect anomaly in data was eliminated by use of index numbers. These index numbers were then used to calculate monthly averages, which were subsequently used for analysis. Different variables changed in different dates thus making it difficult to compare. This anomaly of difference in dates was eliminated by calculating monthly averages. *Thus the data series for all the variables was converted into a monthly time series*. Since GDP data is available on quarterly basis, for some variables the monthly series has been further converted into quarterly series for analysis.

After the initial analysis it was observed that size anomaly is not significant and that absolute and index numbers are giving the same result. Thus subsequent analysis was done using absolute values only.

The research uses basic statistical tools such as correlation, regression and advance statistical tools such co-integration to study the effect and draws conclusion based on the results. The data used is from Indian economy and the time period used is 2011-2014. Monthly and quarterly data has been used as required. Basic statistics of monthly averages for each of the parameter is calculated. Correlation analysis is done amongst selected variables. Linear Regression analysis is done amongst few of the variables based on the research objectives considering mulitcollinearity. Bivariate and Multivariate regression analysis was done using level data as well as data at first difference to account for stationary and autocorrelation in regression. Finally cointegration analysis was done to study the relation amongst time series of variables.

Correlation analysis:

Table 2: Correlations of selected variables using absolute data, monthly series

	CPI	Credit	Borrowings from RBI	Balance with RBI	Total Deposits	USD.Rs	Call Money
CRR	-0.913	-0.950	-0.825	0.834	-0.931	-0.681	0.071
SLR	-0.877	-0.856	-0.820	0.637	-0.860	-0.583	0.157
Repo	-0.643	-0.645	-0.651	0.350	-0.650	-0.762	0.262
ReRepo	-0.658	-0.660	-0.662	0.358	-0.665	-0.762	0.257
BankRate	0.731	0.776	0.645	-0.783	0.747	0.502	0.199

According to Table 2 ,the correlation was found to be high and negative for CRR with Inflation and CRR with Credit off take of commercial banks which clearly indicates towards the 'double edged sword' which a central bank faces. If CRR is reduced, it will increase the credit off take and liquidity in the market but on the other hand it will also increase the inflation level. The correlation between Repo rates and bank borrowings from central bank was negative but medium, indicating that although reducing repo would encourage banks to borrow and increasing liquidity but the effect is not high. Bank rate and call money rates were found to be low on correlation. Bank rate and Deposits with banks were correlated on the higher side indicating that if bank rate is increased, it will increase the deposit rates which will encourage deposits with banks.

Table 3: Correlations of selected variables using absolute data, quarterly series

	GDP	CRR	SLR	Repo	BankRate	Credit	VoM	MD
GDP	1.000	-0.891	-0.858	-0.306	0.758	0.951	0.408	0.941
CRR	-0.891	1.000	0.908	0.477	-0.902	-0.959	-0.072	-0.931
SLR	-0.858	0.908	1.000	0.551	-0.710	-0.894	-0.123	-0.891
Repo	-0.306	0.477	0.551	1.000	-0.248	-0.476	0.411	-0.485
BankRate	0.758	-0.902	-0.710	-0.248	1.000	0.829	0.076	0.787
Credit	0.951	-0.959	-0.894	-0.476	0.829	1.000	0.114	0.993
VoM	0.408	-0.072	-0.123	0.411	0.076	0.114	1.000	0.080
MD	0.941	-0.931	-0.891	-0.485	0.787	0.993	0.080	1.000

Table 3 indicates some interesting correlations about the policy rates. The GDP was found to be negatively correlated with CRR,SLR (High) and repo rates indicating that any increase in these rates by the central bank would negatively affect the economic growth of the country. Also GDP was found to be highly positive correlated with credit levels and money demand, which is a technically and logically correct phenomenon.

Regression Analysis:

The regression analysis was done in bi-variable and multivariate mode using monthly and quarterly time series data at level and at first difference. The regression equations are also given in brackets.

Regression 1: Inflation (CPI) over Cash Reserve Ratio (CRR) at level and on first differences.

The R-squared and DW statistic was found to be 0.83 and 0.19 at level data (CPI=169.71-9.778*CRR) and 0.035 and 0.68 at first difference data, respectively (CPI_d=1.11+0.84*CRR_d).

The two series were found to be not co integrated. Autocorrelation improved but R-squared fell sharply when data set was changed from level to first difference. Regression can be said to be a good fit at level.

Regression 2: Bank Credit over Cash Reserve Ratio

The R-squared and DW statistic was found to be 0.9 and 0.21 at level data (Credit=80047 -6309 * CRR) and 0.17 and 2.47 at first difference data, respectively (Credit_d= 490 -1466.6 * CRR_d).

The two series were found to be co-integrated at first difference. Autocorrelation improved significantly but Rsquared fell sharply. A good model to forecast but unexplained variance remains a concern.

Regression 3: Call Money rates over Bank Rate

The Rsquared and DW statistic was found to be 0.039 and 0.62 at level data (Call Money= 7.4+0.1 * Bank rate) and 0.213 and 1.37 at first difference data, respectively (Call Money_d= -0.05+ 0.47 Bank Rate_d).

The two series are co-integrated at first difference. Rsquared and autocorrelation improved at first level data.

Regression 4: Call Money rate over Repo rates

The R-squared and DW statistic was found to be 0.068 and 0.68 at level data (Call money =4.67 +0.45 Repo) and 0.025 and 1.27 at first difference data, respectively (Call Money_d= -0.015+0.61 Repo_d).

The two series are co-integrated at first difference. Autocorrelation improved for regression at first level.

Regression 5: Foreign exchange rate over CRR

The R-squared and DW statistic was found to be 0.464 and 0.31 at level data (ExRate=74.5-4.15 * CRR) and 0.0238 and 1.6 at first difference data, respectively (ExRate_d= 0.545 + 1.61 * CRR_d).

The two series are co-integrated at first difference. Autocorrelation improved significantly but Rsquared fell sharply.

Regression 6: Credit over Total Deposit

The Rsquared and DW statistic was found to be 0.991 and 0.69 at level data (Credit= -6008.41 + 0.856 * Total Deposit) and 0.17 and 1.94 at first difference data, respectively (Credit_d= 303 + 0.382 * Total Deposit_d).

Not a realistic regression at level as Rsquared is very high at level data. The two series are co-integrated at first difference. Regression is satisfactory and NO autocorrelation was found at first difference. Good fit for forecasting. The regression can be said as a good fit at first difference.

Multivariable regressions:

Regression 7 : Inflation with CRR, SLR, Repo, Re-Repo and Bank Rate

Multicolinearity was expected to effect the regression because of high correlation between CRR and SLR (0.862).CRR and Bank rate (-0.869) and between Repo and Re-repo rates (0.999).

Thus, SLR, Re-repo and Bank rate were reduced from the equation to counter multicollinearity.

At Level: CPI= 187.14 -8.92 * CRR -2.72 *Repo

At first difference: CPI_d= 1.1 + 0.71 CRR_d +0.44 Repo_d

The Rsquared was found to be 0.843 with DW statistic of 0.23 at level. The Rsquared was found to be 0.046 with improved DW statistic at 0.68 at first difference. The series were not co-integrated at level or at first difference.

Regression 8: Credit over Total Deposit , CRR, Repo rate

High correlation was found between Total Deposits and CRR (-0.931) and thus Total Deposit was removed from the regression to remove multicollinearity.

At level: Credit= 88013 -5918 * CRR - 1247 * Repo

At first difference: Credit_d= 485 -1509 CRR_d + 156 Repo_d

The Rsquared was found to be 0.9 with DW statistic of 0.24 at level. Not a realistic regression because of very high Rsquared value.

The R-squared was 0.17 and DW statistic was 2.44 at first difference. The three series were found to be co-integrated at first difference. Good fit model at first difference with satisfactory explained variance as well as LOW autocorrelation.

Regression 9: Call Money Rates over Bank Rate and Repo

At level: Call Money Rate=1.62 + 0.17 Bank Rate + 0.66 Repo Rate

At first difference: Call Money Rates_d= -0.05 +0.47 * Bank Rate_d + 0.6 Repo Rate_d

The Rsquared was found to be 0.16 and DW statistic was found to be 0.67 at level. The Rsquared was found to be 0.23 and DW statistic was found to be 1.34 at first difference. All parameters improved in regression at first level indicating good fit. The three series were found to be co-integrated at first difference. Repo coefficient remained almost constant indicating true coefficient.

Regression of quarterly time series:

Regression 10: GDP over Credit

At level: GDP=2702+0.44 *Credit
The Rsquared was found to be 0.9 and DW statistic was found to be 2.13 at level.

At first difference, the Rsquared was found to be 0.0003 and DW statistic was 2.15 with almost no autocorrelation. The series were co-integrated at first difference but low Rsquared is a concern.

Regression 11: GDP over CRR

At level: GDP= 38670 - 2889 * CRR
The Rsquared was found to be 0.793 at level and 0.014 at first difference . The DW statistic was 1.56 at level and 2.1 at first difference indicating almost no autocorrelation in the regression. The series are also found to be co-integrated at first difference but with a very low Rsquared value.

Regression 12: GDP over Repo

At level: GDP=40990 - 2082 * Repo

The Rsquared was found to be 0.093 at level and 0.29 at first difference. The DW statistic was found to be 0.53 at level and 1.8 at first difference with almost no autocorrelation. The two series were found to be co-integrated at first difference. The series can be said to be a good fit at first difference with 29% explained variance and low autocorrelation.

Regression 13: GDP over CRR and Repo

At level: GDP= 31561 +-3128 * CRR + 1052 * Repo

The Rsquared was found to be 0.811 at level and 0.3 at first difference . The DW statistic was found to be 1.34 at level and 1.78 at first difference with almost no autocorrelation. The two series were found to be cointegrated at first difference. The series can be said to be a good fit at first difference with 30% explained variance and low autocorrelation.

Regression 14: Velocity of Money over CRR

At level: VoM= 131 -0.001 * CRR

The Rsquared was found to be very low at 0.005 at level and 0.06 at first difference . Although, the DW statistic was encouraging with 2.1 at level and 2.25 at first difference but with almost no autocorrelation the series were found to be co-integrated at first difference.

Regression 15: Velocity of Money over Repo

At level: VoM=0.22 +0.012 Repo

The Rsquared was found to be at 0.16 at level and 0.31 at first difference . The DW was 2.04 at level and 2.02 at first difference with no autocorrelation at level or at first difference. The two series were found to be co-integrated at first difference.

Regression 16: Velocity of Money over CRR and Repo

At level: VoM=0.20 -0.004 *CRR + 0.016 * Repo

The Rsquared was found to be 0.26 at level and 0.32 at first difference . The DW statistic was found to be 2.19 at level and 2.02 at first difference with no autocorrelation. The two series were found to be co-integrated at first difference.

Regression 17: Money Demand over CRR

At level: MD= 1.9 -1.9E^6 * CRR

The Rsquared was found to be 0.86 at level and 0.03 at first difference . The DW statistic was 0.75 at level and 1.46 at first difference. Although, autocorrelation improved but Rsquared came down drastically from level data to first difference data. The series were found to be not co-integrated.

Regression 18: Money Demand over Repo

At level: MD= 26210700 - 2110030 * Repo

The Rsquared was found to be 0.23 at level and 0.059 at first difference . The DW statistic was found to be 0.34 at level and 1.21 at first difference. Although autocorrelation improved but Rsquared fell drastically. The series were found to be not co-integrated.

Regression 19 : Money Demand over CRR and Repo

At level: MD= 20557100- 1875740 * CRR -230425 * Repo

The Rsquared was found to be 0.86 at level and 0.02 at first difference . The DW statistic was found to be 0.84 at level and 1.64 at first difference. Although autocorrelation improved but Rsquared fell drastically. The series were found to be not co-integrated.

Conclusion:

The effect of monetary policy changes was observable and was observed but no 'confident' conclusive evidence was found ,although, the results can be used to understand the direction and quantum of effect of monetary policy in limited constraints. For most of the regressions, when autocorrelations were removed, the Rsquared values fell down.

An analysis of basic statistics indicated that CRR and Bank rate emerged as two most widely used rates by RBI as Coefficient of Variation for the two were found to be highest amongst the five policy and monetary rates, also including Repo, Reverse Repo and SLR.

Most of the time series were found to be cointegrated at first difference and with no autocorrelation indicating good fit for forecasting. However, the unexplained variance was a concern in many such cases which needs to be further explored.

The best fit was found between Velocity of Money over Repo and CRR indicating the direct affect of rates and reserves upon Money supply and GDP. Interpretation of the correlation and regression results from the research enhances the understanding of the macroeconomic variables and policy rates and would surely assist in policy making and further research on the subject.

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